

Pediatric Femoral Arterial Cannulations in Extracorporeal Membrane Oxygenation: A Review and Strategies for Optimization

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Venoarterial (VA) extracorporeal membrane oxygenation (ECMO) is a treatment strategy for pediatric patients with cardiopulmonary failure. Femoral cannulation is a commonly used technique for obtaining central access in children and adolescents despite high rates of vascular complications. Ischemic limb complications remain one of the most common problems facing survivors of femoral VA ECMO. Our aim is to review the literature on femoral cannulation in children and to present a review of techniques for femoral cannula placement, decannulation, and arterial repair to help minimize and ameliorate complications related to cannulation. We performed a literature review for pediatric ECMO cannulation, complications, and femoral artery repair in pediatric patients. We focused on the management of arterial and ischemia-related complications during and after ECMO support via femoral cannulation. Vascular complications are not infrequent for patients requiring ECMO support via femoral cannulation. As such, judicious distal perfusion monitoring as well as augmentation of distal flow are important strategies during mechanical support. The use of distal perfusion catheters can be used to minimize the risk of limb ischemia. As femoral cannulation grows more popular in pediatric patients, strategies for arterial repair are essential to ensure extremity perfusion upon decannulation. *ASAIO Journal* 2019; 65:636–641.

Key Words: ECMO, femoral artery repair, femoral cannulation, pediatrics

Extracorporeal membrane oxygenation (ECMO) is a well-established rescue therapy to provide temporary support for children with cardiopulmonary failure. Since its introduction several decades ago, more than 55,000 infants and children have been placed on ECMO internationally.¹ Venoarterial ECMO requires cannulation of a large artery to deliver oxygenated blood to the circulation.^{2,3} In pediatric patients, the carotid artery has traditionally been accessed as the arterial cannulation site when closed-chest ECMO cannulation is performed. This technique typically requires distal ligation of the carotid

artery after cannulation in infants, which has the potential to jeopardize cerebral blood flow. If collateral circulation is not sufficient, patients are at risk for an immediate or subsequent stroke and neurologic or cognitive dysfunction. Studies of neonatal ECMO survivors have shown a comparable incidence of major neurologic disability after carotid ligation to critically ill neonates not requiring ECMO.⁴ However, it is not clear at what age the pediatric brain loses its ability to adapt to ligation of the carotid artery. Regardless, pediatric patients of any age have significantly higher rates of neurologic injury after ECMO via carotid artery cannulation when compared with other strategies,⁵ and as such, alternative methods have been described.

To minimize potential neurologic sequelae from distal carotid artery ligation, the femoral artery may be used for arterial access in appropriate patients. Typically, older toddlers have common femoral arteries large enough to accommodate arterial cannulae of sufficient size to provide adequate inflow support. Studies involving femoral artery catheterization in children less than 5 years of age have shown a high incidence of subsequent arterial occlusion and limb length discrepancies.⁶ It is unclear when the ideal age to transition to femoral cannulation is but based on data from the Extracorporeal Life Support Organization (ELSO) database most centers convert to femoral cannulation by age 5.⁷

As is true with carotid cannulation, femoral arterial cannulation for mechanical circulatory support places patients at risk for disturbances in distal perfusion.⁸ Various techniques have been employed to either prevent or treat threatened limb perfusion, but there are limited resources in the literature describing optimization of perfusion and femoral arterial repair after decannulation in pediatric patients. The purpose of this article is to review femoral cannulation techniques, strategies for optimizing distal perfusion, and repair options for the femoral artery to prevent and ameliorate limb ischemia in this patient population.

Limb Malperfusion During ECMO

The ELSO registry has captured limb ischemia as a complication of ECMO support since 2013. However, rates of limb ischemia reported to the ELSO registry are significantly lower than rates reported in the literature, suggesting that limb malperfusion may be an under-reported complication.⁹ Ischemic complications using various approaches for femoral cannulation have a reported incidence between 10% and 70%. One meta-analysis of 1703 adult patients identified an incidence of limb ischemia in 10% of patients.¹⁰ In the pediatric patient population, reports of limb ischemia are significantly higher, with rates ranging from 29% to 54%.⁹ Regardless of rate, the

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potential for limb perfusion compromise both during and after ECMO support poses a significant threat to pediatric patients and as such, care should be taken to ensure adequate perfusion during support as well as upon decannulation.

Clearly, ensuring adequate perfusion during ECMO support is critical, and improvements in strategies to optimize perfusion are warranted. Multiple strategies have been described to augment distal perfusion utilizing antegrade and retrograde catheters during mechanical support.^{11,12} At this time, there are no accurate methods to predict which patients will develop significant limb ischemia. Furthermore, the use of these antegrade and retrograde perfusion catheters is limited by technical challenges in pediatric patients. Not only is lower extremity perfusion potentially impaired while on mechanical support, but also, upon decannulation, femoral artery injury or thrombosis may threaten distal limb perfusion. Vascular injury after ECMO decannulation presents numerous surgical challenges to achieve a successful repair and restore adequate limb perfusion. Management of lower extremity vascular compromise secondary to femoral artery injury or occlusion after decannulation is inadequately described in the pediatric patient population. Techniques for cannulation, distal perfusion, perfusion monitoring, decannulation, and vascular repair will be reviewed.

Femoral Cannulation Techniques

There are multiple methods for femoral artery cannulation described in the literature for children, including open, percutaneous, and hybrid techniques. The rates of lower extremity ischemia during ECMO in pediatric patients appear to be independent of the technique employed for cannulation. Regardless of technique, a heparin bolus is administered before arterial access. There is no consensus on the appropriate dose before pediatric ECMO cannulation. In adults, a 5,000 Unit intravenous bolus is typically accepted as an appropriate initial dose. In pediatric patients at our institution, we deliver a 100 U/kg dose of heparin before cannulation. For patients considered at high risk of having hemorrhagic complication, we use a lower dose of 75 U/kg. After this bolus, a heparin infusion is initiated when the activated clotting time (ACT) drops below 300 seconds, and this infusion is titrated based on ACT with a goal of 180–220 seconds.⁹

If femoral venous access is obtained in addition to femoral arterial access, it is prudent to attempt to obtain venous access on the opposite side of arterial inflow for two reasons. First, this has been documented to minimize vascular compromise of the extremity from impaired venous return.⁹ In addition, this allows for more complete closure of the groin wound once the cannulas are in place in the case of open cannulation. The smallest arterial cannula that allows sufficient cardiopulmonary support should be used. Full flow for pediatric patients can be estimated by calculating the product of body surface area and a cardiac index or 2.4 L/min/m². The smallest cannula capable of delivering adequate flow and be accommodated by the patient should be selected for placement.⁹ A summary of cannula sizing by patient size is provided in **Table 1**.

Open Technique

Open femoral arterial cannulation allows for visualization of the vessels to ensure adequate size, direct placement of a purse

Table 1. Sizing Chart for Arterial and Venous ECMO Cannulae in Pediatric Patients

Patient Size (kg)	Arterial Cannula (Fr)	Venous Cannula (Fr)
2	8	8–10
3–6	10	10–12
6–8	12	14
8–16	14	17
16–30	17	19
30–40	17	21
>40	21	25

ECMO, extracorporeal membrane oxygenation.

string or graft, confirmation of cannula position, and hemostasis.¹ A longitudinal incision is made just below the inguinal crease. The common femoral artery and vein are isolated, and proximal and distal control is obtained with vessel loops. After heparinization, a concentric purse string is placed in the vessel. Of note, the purse string should be placed in a long and narrow manner as this has been described as a potential method to avoid stenosis of the vessel.¹ The cannula is inserted after a longitudinal arteriotomy is created, and the purse string is tightened around the cannula to ensure hemostasis.

The chimney technique is described as an effective cannulation strategy in patients with small femoral vessels. This technique utilizes a polytetrafluoroethylene or Dacron “chimney graft” that is anastomosed to the vessel during open exposure (**Figure 1**). These grafts offer lower arterial line pressures and decrease the risk of malperfusion of the lower extremity distal to the cannulation site.¹

Percutaneous Technique

To employ a percutaneous cannulation technique, ultrasound should first be used to confirm adequate vessel size to

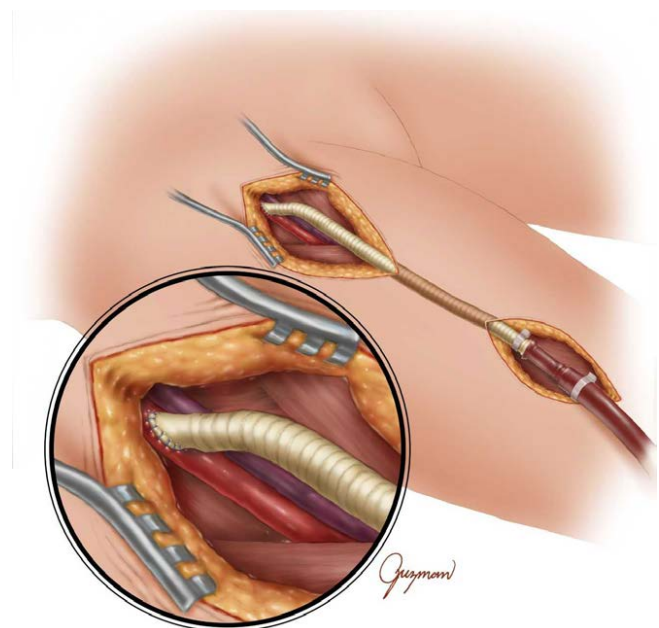


Figure 1. Femoral arterial cannulation via the chimney technique.

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avoid vessel transection and extremity ischemia when possible. After aseptic groin prep applied, the common femoral artery is accessed with a large bore needle, and a guide wire is passed. Once wires have been placed and secured, anticoagulation is administered. The Seldinger technique is then employed. A stab incision is made at the base of the wire, and a series of dilators are serially passed in increasing size until the arterial cannula can be inserted.¹ The arterial cannula is advanced to the level of the external iliac artery taking care not to advance cannula into the aorta and potentially occluding blood flow to the contralateral extremity. Care must also be taken when advancing the cannulas around the angle of the pelvis to prevent accidental vascular injury. A purse string can be used at the level of the skin to prevent bleeding around the cannula once access has been obtained.

Hybrid Technique

For this open Seldinger technique, a small vertical or transverse incision is made in the groin with dissection carried down to the anterior surface of the vessels. A concentric purse string is placed, typically with 5-0 Prolene suture, again with careful consideration taken to make the purse string long and narrow so as to avoid stenosis of the vessel.¹ The vessel is then accessed, and a cannula is placed in a similar fashion to the percutaneous Seldinger technique but under direct vision. The purse string is tightened, the cannula is secured, and groin incision is closed around the cannula.

Distal Perfusion Catheterization

Numerous strategies can be employed to establish or improve distal perfusion during femoral cannulation. Distal perfusion can be accomplished *via* antegrade flow with a catheter in the superficial femoral artery (SFA) or *via* retrograde flow perfusion with a catheter introduced into the dorsalis pedis or posterior tibial arteries.³ Antegrade distal perfusion catheter (DPC) placement is performed using a 4–8 French single lumen sheath placed under ultrasound guidance in the SFA (**Figure 2**). It is preferable to achieve wire access of the SFA before insertion of the arterial cannula in the common femoral artery to visualize flow using ultrasound. When ultrasound guidance fails, a cut down and puncture under direct visualization can be performed.⁹ The DPCs can then be connected *via* extension tubing to the arterial limb of the ECMO circuit to provide antegrade flow to the limb.

Clinicians can place antegrade SFA perfusion cannulae either prophylactically or after observation on the ECMO circuit to assess for limb viability. Small studies have shown that prophylactic cannulation of the SFA for distal perfusion may reduce rates of ischemia-related complications although this has been shown primarily in adult patients.^{9,11,13} Foley *et al.*¹³ reported that distal ischemia occurred in 21% of patients with femoral cannulation without prophylactic SFA cannulation. Half of these patients required subsequent SFA cannulation to restore perfusion. However, DPCs are not without complications, particularly in pediatric patients. Placement poses risk of SFA injury, SFA thrombosis or distal embolism, and bleeding¹⁴ which place patients at further risk of limb ischemia. Alternatively, retrograde DPCs can be placed *via* percutaneous

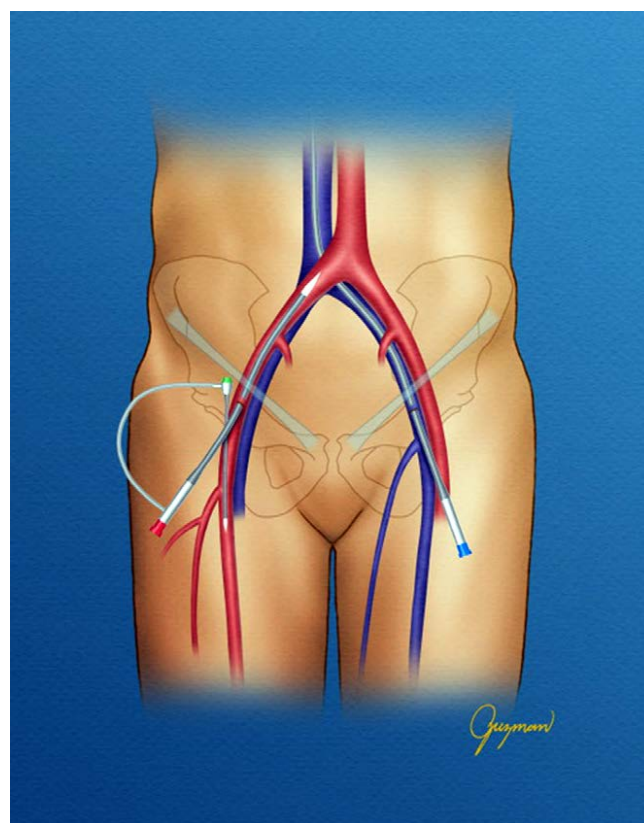


Figure 2. Extracorporeal membrane oxygenation femoral cannulation. Arterial and venous cannulae are placed in the femoral artery and vein as depicted in the illustration. A distal perfusion catheter is placed in the femoral artery distal to the arterial cannula position to provide flow to this extremity. [full color online](#)

insertion of a small caliber catheter into the dorsalis pedis or posterior tibial artery for limb perfusion.¹⁵

Hendrickson *et al.*¹⁶ instituted a distal perfusion strategy in patients undergoing femoral cannulation after reporting 11.5% of patients undergoing femoral cannulation experienced complications related to malperfusion when distal perfusion was not established at the time of cannulation. After employing a distal perfusion strategy, the authors reported no further episodes of lower extremity ischemic complications.¹⁶ In this study, authors describe an open technique for femoral cannulation in which they insert their proximal cannula and distal perfusion cannula *via* the same arteriotomy, utilizing Rummell tourniquets to ensure no back bleeding around either cannula. This potentially creates a larger arteriotomy that will require subsequent repair.

Schad *et al.*⁹ performed a single-center, retrospective review of pediatric patients undergoing ECMO support, comparing those patients who had prophylactic DPC to those who had DPCs placed in response to clinically evident limb malperfusion. In their series of 29 pediatric patients, 12% received prophylactic DPC while 29% required reactive DPC placement for ischemic complications of femoral cannulation.⁹ Importantly, and in contrast to Foley *et al.*¹³, who reported no ischemic complications among adult patients who received prophylactic SFA DPC placement, in this series of pediatric patients, prophylactic DPCs did not eliminate limb ischemia. There was a statistically insignificant difference in development

of clinically significant limb ischemia in patients who underwent prophylactic *versus* reactive DPC catheter placement.⁹ Of note, DPCs placed reactively for clinical ischemia did not eliminate subsequent progression to surgical intervention in this series.

Although their study is limited by small sample size and retrospective methodology, Schad *et al.*⁹ elucidate an important point in pediatric patients; that is to say that femoral cannulation in these patients frequently results in distal malperfusion, regardless of prophylactic measures taken to mitigate this risk. With this in mind, development of effective methodologies for both perfusion monitoring as well as restoration of distal perfusion and femoral artery repair are critical in these patients to prevent morbidity resulting from lower extremity ischemia.

Monitoring Distal Perfusion During ECMO

Once cannulation has been achieved and ECMO support has been initiated, prudent monitoring of lower extremity perfusion is critical to preventing ischemia-related injury to the leg. While on mechanical support, swift recognition of impaired distal perfusion is critical so that rapid intervention can be achieved. At our institution, hourly checks for capillary refill and Doppler signal checks help alert the team to acute changes in perfusion. Some have reported protocols utilizing continuous measurement of oxygen saturation of the toe and comparison with upper extremity oxygen saturations.¹⁷ In addition to clinical exam and Doppler signal checks, other institutions have utilized noninvasive flow probes applied to DPC connection tubing as well as utilization of near-infrared spectroscopy for perfusion monitoring.⁹

In cases of acute limb ischemia during ECMO, it is crucial to identify the cause to be either low flow state or embolism.¹⁷ Ischemia because of poor arterial flow secondary to obstruction of the cannula, as is commonly the cause in the pediatric population, can be prevented by prophylactic placement of DPCs or via the chimney cannulation technique as previously mentioned. Distal embolism requires intervention to remove the occlusive clot.

Femoral cannulation introduces risk not only for distal lower extremity ischemia but also furthermore, cerebral, upper extremity, and cardiac perfusion may be impaired secondary to a phenomenon known as the Harlequin syndrome. In the so-called cool head, warm legs syndrome, the arteries arising from the ascending and aortic arch are supplied by a mixing of deoxygenated blood via native left ventricle with highly oxygenated blood from the ECMO circuit via the femoral artery cannula.¹⁸ The result is upper extremity and possibly cerebral hypoxia secondary to deoxygenated blood being delivered to the upper body arterial circulation. One retrospective study of 159 patients undergoing ECMO via femoral percutaneous cannulation reported an incidence of upper body hypoxia in 8.8% of patients.¹⁸ Harlequin syndrome is an oxygenation issue and can be managed with placement of an additional jugular vein cannula, creating what is known as veno-arterial-venous (V-AV) ECMO. The V-AV ECMO allows for variable flow of oxygenated blood through the femoral arterial and internal jugular venous cannulas to provide oxygenated blood to the upper extremities and head. This configuration can pose challenges in flow management, and often patients will be transitioned to veno-venous ECMO once the heart is fully recovered.¹⁸

Femoral Arterial Decannulation Techniques

Open Technique

Removal of cannulae placed via the open technique is best performed in the operating room. The operative field is reopened, and the cannulation sutures are re-evaluated. Proximal and distal arterial control is obtained. With smaller cannulae, a purse string suture can be used to close the arteriotomy. Closure of the arteriotomy can lead to vessel stenosis, thrombosis, or embolism and may compromise distal perfusion. In those cases where arterial stenosis or distal pulselessness occurs, arterial reconstruction is required, as is described below, to restore perfusion.

Percutaneous Technique

In cases of percutaneous cannulation, an open repair in the operating room can be performed as previously mentioned. Newer percutaneous techniques describe a bedside method. After 5 to 10 minutes of manual compression, a femoral compression system (FemoStop II Plus; Radi Medical Systems, Uppsala, Sweden) can be applied to create mechanical pressure over the vessel puncture to achieve hemostasis. The pressure of the pneumatic dome of the device is controlled by a reusable pump with manometer and remained in position over the arterial puncture site for at least 10 hours. When this device is used, prudent observation is necessary to ensure both hemostasis and distal perfusion.

Femoral Artery Repair

Decannulation presents a variety of potential complications that place patients at risk for morbidity and mortality regardless of the cannulation site. When femoral vessels are accessed, potential complications after decannulation include femoral artery thrombosis, lower extremity thromboembolism, pseudoaneurysm formation, arterial dissection or transection, and arteriovenous fistulae. Resulting lower extremity ischemia and potential for reperfusion injury with resultant compartment syndrome pose significant threats to patients. These complications occur in both adults and pediatric patients and require prompt surgical intervention to prevent critical limb ischemia.

In pediatric patients, femoral arterial complications after ECMO decannulation pose therapeutic challenges to maintain or restore adequate limb perfusion. Although interventions to restore distal limb perfusion have been well described in adult patients,¹⁷ surgical management of interruption of lower extremity perfusion after ECMO decannulation has not been well described in pediatric patients. Open femoral arterial repair is often required in cases of compromised limb perfusion. When surgical repair is necessary, we utilize ipsilateral saphenous vein to create a vein patch for arterial repair (**Figure 3**). Saphenous vein is dissected, with care taken to suture ligate branches to ensure hemostasis, and is ligated distally and proximally. The harvested vein is then fashioned into a patch to be placed at the arterial repair site and is sutured in place with 5-0 Prolene suture once flow has been restored (**Figure 3**).

After arterial repair, medical management and imaging surveillance are advised to ensure no further complications develop. There is no consensus in the literature currently on

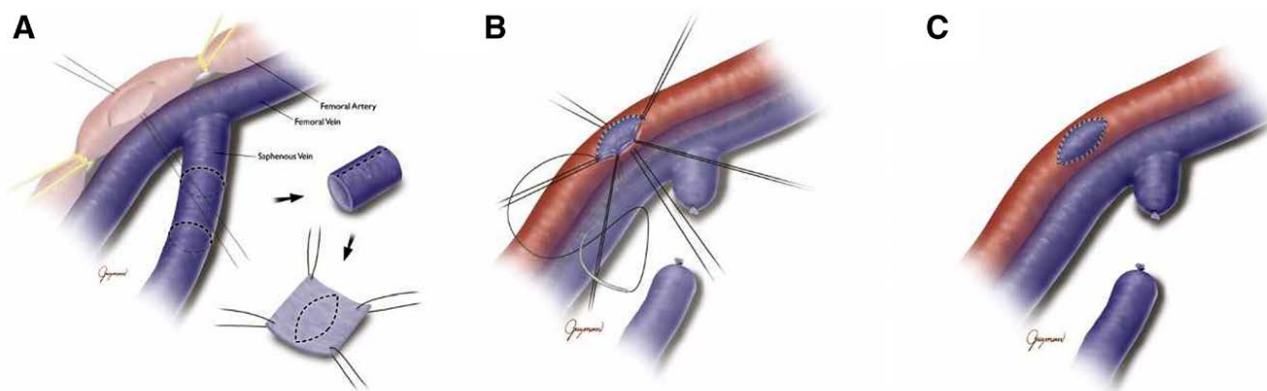


Figure 3. Femoral artery repair using ipsilateral saphenous vein. **A:** The saphenous vein is harvested for creation of a vein patch for the arterial repair. The femoral artery is dissected, proximal and distal control is achieved, and a longitudinal arteriotomy is created for repair. **B:** The saphenous vein patch is anastomosed in a running fashion over the arteriotomy once the perfusion deficit has been resolved. **C:** The completed femoral artery repair with saphenous vein patch. [full color online](#)

postoperative or long-term management for children after lower extremity arterial repair. Postoperatively, we continue Doppler signal checks hourly for at least 24 hours after repair and then every 2 hours for at least a week after the arteriotomy has been repaired. At our institution, we recommend maintaining these patients on acetylsalicylic acid for 3–6 months. We also recommend duplex ultrasonography to confirm vascular patency every 3–4 months for the first year after repair and then every 6 months for the second year after repair.

Discussion

Multiple studies have demonstrated the importance of the antegrade DPC to minimize complications during ECMO. This can be placed by both percutaneous and open techniques. The appropriate flow for arterial support in the distal artery is unclear. In adults, the flow in the common femoral artery under normal circumstances is approximately 350 ± 141 mL/min.¹⁹ Flows within the DPC when the overall circuit is running at 4 L/min are generally around 240 mL/min, which appear to be adequate to support limb perfusion in our experience.

The greatest challenge for the prevention of femoral vascular complications during ECMO is the rapid clinical diagnosis of embolism or cannula-dependent vessel occlusion. Pediatric patients on mechanical support present numerous clinical challenges that prevent the detection of complications. These include the difficulty in detecting peripheral pulses, the patient's sedation status, high levels of circulating catecholamines causing peripheral vasoconstriction, and generalized edema.²⁰ Duplex ultrasounds can be used to document patency and adequate distal perfusion when in doubt. In cases where perfusion is inadequate and arterial repair is ineffective, the assistance of vascular surgery may be beneficial for femoral artery bypass or other strategies to augment flow.

Nursing protocols have been designed to monitor for vascular complications while on venoarterial (VA) ECMO. These generally involve hourly clinical examination of the feet and legs for temperature, color, capillary refill, and compartment syndrome. In addition, peripheral arterial perfusion monitoring with hourly Doppler signals and routine biochemical examination of myoglobin and creatinine kinase can help alert the physician to an impending ischemic complication. In cases

of ambiguity, arterial duplex ultrasonography or contrast-enhanced computed tomography angiography can readily establish the adequacy of arterial flow.

As described above, swift intervention to re-establish distal flow is necessary to reduce malperfusion-related morbidity. Placement of DPCs can be an effective strategy to reduce limb ischemia. However, DPCs have not been shown to entirely eliminate the risk of limb ischemia in pediatric patients undergoing femoral cannulation. Vascular injury, thrombosis, and embolism can all occur as a result of femoral arterial access. Repair strategies must be in place in the event of these complications to ameliorate disruptions in limb perfusion. Early involvement of vascular surgeons should be considered when complex repairs or vessel bypasses are required to achieve adequate distal perfusion both during and after ECMO support.

Conclusions

Femoral cannulation for pediatric patients who require VA ECMO support is an effective technique when utilized in appropriate patients with distal perfusion optimization and prudent monitoring. Utilizing femoral vessels for ECMO cannulation when appropriate can mitigate the risks of carotid artery cannulation while also avoiding the morbidities associated with central cannulation. However, strategies to optimize distal perfusion and femoral artery repair must be employed to minimize the risks of limb ischemia in pediatric patients requiring mechanical support.

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